

For time-axis, whichever (crest) is behind is leading.

- 17 B** As α particle moves near nucleus, it experiences an electric force of repulsion that causes it to slow down to zero velocity. The distance r is known as the distance of closest approach.

$$\begin{aligned}\text{loss in ke} &= \text{gain in pe} \\ &= Qq / 4\pi\epsilon_0 r \\ 4.9 \times 10^6 \times 1.60 \times 10^{-19} \times 10^6 &= (2 \times 1.6 \times 10^{-19}) (79 \times 1.6 \times 10^{-19}) / 4\pi \times 8.85 \times 10^{-12} r \\ r &= 4.64 \times 10^{-14} \text{ m}\end{aligned}$$

